

RESHME-NAMMA PRIDE

Project ID: 52

Android App Development using Generative AI

Agriculture — Sericulture Guard Application

“Silk is Our Pride” - A Digital Supervisor for Farmers

MindMatrix VTU Internship Program

Product Requirements Document (PRD)

Version 1.0

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1 Executive Summary

Project Overview

Karnataka stands as the undisputed Silk Capital of India, contributing significantly to the nation's agricultural economy. However, the backbone of this industry—the silkworm—is biologically fragile. **Reshme-Namma Pride** is a pioneering Android application designed to democratize technology for small-scale sericulturists. By replacing expensive IoT infrastructure with intelligent algorithmic logic, this app serves as a “Digital Supervisor,” ensuring optimal yield and minimizing crop failure risks.

» The Critical Challenge

Sericulture is highly sensitive to climate. The project aims to solve the critical issue of environmental monitoring where even a slight deviation can cause massive economic loss. Unlike modern large-scale agriculture, small farmers often lack access to sophisticated tools, relying instead on generational intuition which is susceptible to modern climate volatility.

⚠ Critical Pain Points

Why is this critical?

- **Sensitivity:** A fluctuation of just 2°C can induce “Grasserie” disease or kill cocoons instantly.
- **Cost:** IoT sensors cost $\approx 10,000+$, unaffordable for marginal farmers with small land-holdings.
- **Knowledge Gap:** Generic advice fails; Instar-specific monitoring is required for each growth stage.
- **Risk:** Crop failure risk is high with no early warning systems, leading to debt traps for farmers.

» The Solution

The application uses a “Human-in-the-loop” approach. The farmer acts as the sensor (reading basic thermometers), while the App acts as the Brain (processing data via GenAI logic). This hybrid model ensures affordability while maintaining high-tech precision.

2 Project Vision & Goals

Reshme-Namma Pride — meaning “Silk is Our Pride” — is envisioned as a Sericulture Guard. It empowers farmers to make data-driven decisions using a GenAI-powered logic engine, without needing internet connectivity.

Attribute	Detail
App Name	Reshme-Namma Pride (Sericulture Guard)
Target User	Small-scale silkworm farmers in Karnataka and South India.
Platform	Android (offline-first), supporting Android 8.0+ devices.
Core AI	GenAI Logic Engine — stage-aware advice per Instar using decision trees.
Database	Room DB — local, batch-wise persistent storage for offline reliability.

Mission Statement

“To provide every farmer with a digital supervisor that fits in their pocket, making complex biological monitoring simple, affordable, and actionable. We aim to bridge the digital divide in Indian agriculture.”

3 User Flow & App Usage

The application follows a simple 5-step workflow designed for ease of use, specifically tailored for users who may not be tech-savvy. The UI prioritizes large buttons, high contrast text, and local language support.

» Detailed Step Descriptions

1. Step 1: Batch Tracker

The farmer initializes the cycle by entering the start date and the specific silkworm breed (e.g., PM x CSR2). The app automatically calculates the expected harvest date and sets the initial Instar stage to 1.

2. Step 2: Climate Entry

The user is prompted three times a day (Morning, Afternoon, Evening) to input Temperature and Humidity. The input interface enforces validation to prevent impossible entries (e.g., Temp range: 10°C–45°C).

3. Step 3: Smart Advice

The Logic Engine compares the current inputs against the “Ideal Growth Curve” for the specific Instar stage. It generates text advice in simple language, such as: *“Temperature is high! Open windows and spread wet gunny bags on the floor.”*

4. Step 4: Climate Dial

A visual circular dial updates in real-time on the home screen:

- **Green**: Safe Zone — Continue regular maintenance.
- **Orange**: Caution Zone — Monitor closely for changes.
- **Red**: Danger Zone — Immediate action required.

5. Step 5: Harvest Alert

Based on the lifecycle duration, the app sends a notification when it is time to transfer mature worms to the spinning trays (Mounting), preventing over-ripening.

4 Scope of the Application

» In Scope Features

The following features are critical for the Minimum Viable Product (MVP) v1.0 release.

- ✔ **Batch Management:** Full CRUD (Create, Read, Update, Delete) operations for tracking silkworm batches by date and breed.
- ✔ **Manual Climate Entry:** Intuitive forms for Temperature & Humidity logging 3 times daily.
- ✔ **Instar Logic Engine:** Stage-aware (1–5) advice using if-else/when logic blocks.
- ✔ **Visual Indicators:** Color-coded Climate Dial for at-a-glance status checking.
- ✔ **Offline Storage:** Room DB persistence to ensure data is never lost.
- ✔ **History Logs:** Ability to view past batch data for trend analysis.

» Out of Scope Features

Features planned for future versions or explicitly excluded from v1.0.

- ✘ **IoT Integration:** Direct connection to Bluetooth or WiFi sensors is not supported in v1.0 to keep complexity low.
- ✘ **Cloud Sync:** No backend server or cloud storage; data remains strictly local.
- ✘ **Financials:** Sales tracking, cost calculation, and market price integration are excluded.
- ✘ **iOS Version:** The application is built natively for Android only.

5 Functional Requirements

ID	Requirement	Description & Acceptance Criteria
FR-01	Batch Management	Users can create a new batch profile. <i>Criteria:</i> Auto-generated ID, Date picker, Breed dropdown.
FR-02	Climate Data Entry	Input form for Temp/Humidity. <i>Criteria:</i> Reject values < 10°C or > 45°C. Toast error for invalid data.
FR-03	Instar Tracking	System auto-calculates Instar stage. <i>Criteria:</i> Logic based on Start Date (e.g., Days 1-3 = Instar 1).
FR-04	Advice Engine	GenAI Logic triggers on save. <i>Criteria:</i> Advice text must change based on the calculated Instar.
FR-05	Climate Dial	UI Component renders status. <i>Criteria:</i> Green/Orange/Red color state changes instantly.
FR-06	Harvest Timer	Background service checks date. <i>Criteria:</i> Notification triggers on Day 25 (or breed-specific maturity).
FR-07	History Log	List view of entries. <i>Criteria:</i> Recyclerview displaying timestamp, temp, humidity, and advice given.

» Non-Functional Requirements

- **Performance:** App shall launch in under 3 seconds on entry-level devices. Advice generated within 500ms.
- **Usability:** Text size must be large (16sp+) for outdoor readability. High contrast themes required.
- **Reliability:** 100% offline capability. No crash on data null.
- **Security:** No external permissions (Camera/Location) required.

6 Technical Implementation

» System Architecture

The application is built using the **MVVM (Model-View-ViewModel)** architecture pattern. This ensures separation of concerns, making the app easier to test and maintain.



» The Logic Engine

The core intelligence is a decision matrix implemented via Kotlin **when** expressions.

Instar	Ideal Temp	Tolerance	Action if High
1 & 2	28°C	±1°C	Spray water, spread wet gunny bags.
3 & 4	26°C	±2°C	Increase ventilation, use fans.
5	24°C	±2°C	Reduce humidity, dry environment.

i Technical Insight

Database Schema:

- **BatchTable:** ID (PK), StartDate, BreedName, CurrentInstar.
- **LogTable:** LogID (PK), BatchID (FK), Timestamp, TempValue, HumidValue, Advice-Text.

7 Feature List

Must-Have Features (v1.0)

- ✓ Batch creation with date/breed.
- ✓ Manual Temp & Humidity entry.
- ✓ Input validation (10°C–45°C).
- ✓ Instar stage auto-tracking.
- ✓ Color-coded Climate Dial.
- ✓ Offline-first Room DB storage.
- ✓ Harvest Timer alerts.
- ✓ History view per batch.

Good-to-Have Features (v1.1+)

- + Kannada language localization.
- + Push notification reminders.
- + Graphical climate trend charts.
- + Export batch report as PDF.
- + Dark mode support.
- + Voice input for data entry.
- + Multi-farm management.
- + Disease identification guide.

8 Success Criteria for Students

✓ Internship Evaluation Checklist

1. **Instar-Aware Logic:**

The app **must** change its advice based on the Instar stage. Test: Input 25°C. Verify that the advice for Instar 1 is different from Instar 5.

2. **Input Range Enforcement:**

The Climate Entry screen must reject any temperature input outside **10°C–45°C**. Clear error messages (Toast/Snackbar) must be displayed.

3. **Actionable UI:**

The UI must prioritize clarity. The Climate Dial must be visible on the main screen and update instantly upon data entry.

4. **Data Persistence:**

All batch and climate data must survive app restarts and device reboots. Test: Force stop app, clear from memory, reopen—data must be intact.

5. **Offline Functionality:**

The app must be fully functional **without** any internet connection. No network calls should be made.

Impact Goals

Economic Stability: Reduces risk of crop failure, saving farmers thousands of rupees per season.

Global Leadership: Maintains Karnataka's position as a premium silk producer by ensuring quality.

Agri-Tech for All: Makes complex biological monitoring accessible to all, regardless of education level.

9 Glossary of Terms

Term	Definition
Sericulture	The rearing of silkworms for the production of raw silk.
Instar	A stage between molts in the development of an insect larva. Silkworms have 5 instars.
Rearing House	A dedicated room or shed where silkworms are raised under controlled conditions.
Mounting	The process of transferring mature larvae to spinning frames (chandrakis) to spin cocoons.
Room DB	A SQLite object mapping library for Android that provides local data persistence.
MVVM	Model-View-ViewModel; a software architectural pattern that separates the user interface from the business logic.
Grasserie	A common silkworm disease often caused by high temperature and humidity fluctuations.

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